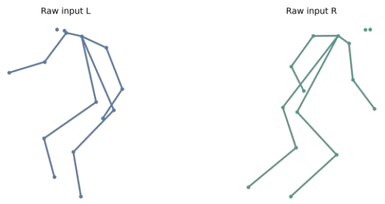
Feature construction and reliability-gated fusion

1 Input pose streams



P^L, P^R
$T \times J \times 3$
per stream

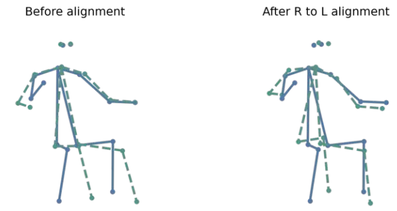
2 Human-centric canonicalization



Hip midpoint, hip axis,
and neck direction
First-frame body reference

Canonical poses: $P^{\mathcal{L}}$, $P^{\mathcal{R}}$

3 Sequence-level alignment



Both directions
$R \rightarrow L$ / $L \rightarrow R$
One similarity fit
per sequence

4 Five geometric-motion cues · Left stream shown; swap L and R for the right stream

Own-view pose



$P^{\mathcal{L}}$

3 ch

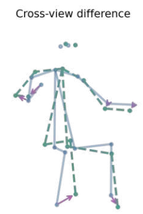
Aligned other view



$P^{\mathcal{R}\rightarrow L}$

3 ch

Cross-view difference



$D^L = P^{\mathcal{R}\rightarrow L} - P^{\mathcal{L}}$

3 ch

Temporal velocity



$V^L_t = P^{\mathcal{L}}_t - P^{\mathcal{L}}_{t-1}$

3 ch

Alignment rotation



$r^{R\rightarrow L}$ (broadcast)

3 ch

Concatenate → X^L, X^R | 3 + 3 + 3 + 3 + 3 = 15 ch per joint and frame

5 Shared joint encoder

$15 \rightarrow 128$
MLP · LayerNorm · GELU

$H^L = \varphi(X^L)$, $H^R = \varphi(X^R)$
Shared parameters

6 Bidirectional cross-view attention

$$\hat{H}^L = \mathrm{LN}(H^L + \mathrm{Attn}(H^L, H^R, H^R))$$

$$\hat{H}^R = \mathrm{LN}(H^R + \mathrm{Attn}(H^R, H^L, H^L))$$

Attn(Q, K, V) · Joint attention per frame

7 Temporal refinement



t−2 | t−1 | t | t+1 | t+2

Depthwise temporal convolution, k = 5

8 Joint-frame gating and residual correction

Base candidates from the canonical and aligned poses

$B^L = \frac{1}{2}(P^{\mathcal{L}} + P^{\mathcal{R}\rightarrow L})$
$B^R = \frac{1}{2}(P^{\mathcal{R}} + P^{\mathcal{L}\rightarrow R})$



$P_{base} = \alpha B^L + (1 - \alpha) B^R$

$P^F = P_{base} + \Delta P$

α

Gate head + Sigmoid

$\alpha = \sigma(g_\theta(Z))$

$\alpha \in [0, 1]^{T\times J\times 1}$

Joint-frame weights
α for B^L; 1 − α for B^R

$Z = [H^L, H^R]$

Residual head → ΔP

Self-collected A01 / pro_1 / frame 746 · Qualitative example · No ground truth